Fraction Models: Are the Parts Equal?
Answer Key

Grade 2–3 Mathematics

Quick Answers

1. $\frac{1}{4}$
2. $\frac{1}{6}$
3. —
4. $\frac{3}{8}$
5. —
6. 4
7. 3
8. —

Curriculum

Level: Grade 2–3 Mathematics
3.NF.A – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

2. *If they got 0.33: counted the 3 unequal pieces as if they were thirds*
4. *If they got 0.33: counted 1 piece out of 3 unequal pieces instead of the equal small parts*
6. *If they got 5: folded after 3 squares and called the leftover 5 squares the other half*
7. *If they got 4: took the largest piece (the 4 squares left after breaking off 2 and 3) as one share*

1. Correct model — name the fraction.

Step 1. Check the parts. The rectangle is cut by three lines into four strips of the same width, so the parts *are* equal. The answer to the first question is *yes.*
Step 2. Because the parts are equal, the model does name a fraction. The bottom number is how many equal parts make the whole (4); the top number is how many are shaded (1). $\boxed{\frac{1}{4}}$
Teaching note: this is the only model on the sheet that is already correct. Students who answer "no" to the first question are usually looking at the shading, not the part sizes.

2. Find and fix — Ben's strip.

Step 1. Ben counted *pieces*, not equal parts. His three pieces are 1, 2 and 3 small parts wide, so they are not the same size and "1 out of 3 pieces" does not name a fraction of the strip.
Step 2. Use the equal small parts instead. The strip is made of 6 equal small parts, and the shaded piece covers 1 of them. $\boxed{\frac{1}{6}}$
Common wrong answer: $\frac{1}{3}$ — one piece out of three unequal pieces. Ask the student to lay the shaded piece on top of the widest piece: it clearly does not cover it, so the pieces cannot all be thirds.

3. Manual review — Kim's pizza (drawing + explanation).

(a) Expected sentence. Any answer with this idea: *the four pieces are not the same size, so no single piece is a fourth.* The shaded wedge is the smallest of the four, so it is less than $\frac{1}{4}$ of the pizza. Do not accept "because there are four pieces" with no mention of size.
(b) Expected drawing. A circle cut by two straight lines through the centre, at right angles to each other, giving four wedges of the same size, with exactly one wedge shaded. Accept any four congruent wedges; the cuts do not have to be horizontal and vertical.

4. Find and fix — Nina's strip.

Step 1. Same mistake as Ben's, one step harder: Nina's three pieces are 3, 2 and 3 small parts wide. Unequal pieces cannot be thirds.
Step 2. Count in equal small parts. The whole strip is 8 equal small parts, and the shaded piece covers 3 of them. $\boxed{\frac{3}{8}}$

Common wrong answer: $\frac{1}{3}$. Note that here the shaded piece is one of the *larger* pieces, so the wrong answer $\frac{1}{3}$ is actually smaller than the truth — guessing from the picture will not rescue it.

5. Comparing two unit fractions of the same whole.

Step 1. The strips are the same length, so we are comparing parts of the same whole — that is what makes the comparison fair.
Step 2. Strip A is cut into fewer equal parts than Strip B, so each of its parts is longer. More equal parts means smaller parts.

$$\boxed{\frac{1}{4} > \frac{1}{6}}$$

Teaching note: the picture is the proof. Students who answer $<$ are reading "6 is bigger than 4" straight off the denominators.

6. Find and fix — Tara's halves.

Step 1. Tara's fold makes pieces of 3 squares and 5 squares. Halves must be the same size, and 3 and 5 are not the same, so this model does not show halves.
Step 2. Share the 8 equal squares between 2 equal halves: $8 \div 2 = 4$. The fold line belongs after the fourth square. $\boxed{4}$
Common wrong answer: 5 — the number of squares left over after Tara's fold.

7. Find and fix — Cal's fair share.

Step 1. Cal's pieces are 2, 3 and 4 squares. A fair share means every friend gets the same amount, so pieces of three different sizes are not a fair share.
Step 2. There are 9 equal squares and 3 friends, so each share is $9 \div 3 = 3$ squares. Each friend then has $\frac{1}{3}$ of the bar, because the bar has been split into 3 equal shares. $\boxed{3}$
Common wrong answer: 4 — the size of Cal's largest piece.

8. Manual review — Zoe's claim (explanation + drawing).

(a) Expected sentence. Any answer carrying this idea: *cutting into four pieces is not enough — the four pieces must also be the same size.* In Zoe's picture the shaded piece is much narrower than the others, so it is less than a fourth of the rectangle. Do not accept "she counted wrong".

(b) Expected drawing. The same rectangle cut into four parts of equal width (three cuts, evenly spaced), with exactly one part shaded. A quick check a student can do themselves: each part must be able to cover any other part exactly.

Teaching note: this is the whole worksheet in one problem. The number of pieces gives the bottom number of a fraction only after the pieces have been made equal.